FIN 3080 Investment Analysis and Portfolio Management

Spring 2026

CUHK (SZ)

Assignment 2

Due: 23:59, March 29, 2026

Disciplines:

- **Late submissions** are **not allowed** for any reason, whether due to physical issues or technical difficulties. A **penalty of 0.5 points** will be deducted for each day late. Submissions made **after the answers** have been posted will not be **counted**.
- A complete submission must include:
  - One readable PDF (Times New Roman, 1.5-spaced, 11pt font, not exceeding 5 pages, named ***YourID_YourName_ResultsPDF.pdf***) containing arguments, tables, and figures. **You can write the report for this assignment either in English or in Chinese.**
  - A compressed archive named *YourID_YourName.zip* with all code files (also named like ***YourID_YourName_Q1Code***) needed to reproduce empirical results.
- Collaboration with peers is permitted, but plagiarism or data fabrication will result in disciplinary action. If two or more submissions are found to be similar, all involved parties will receive **0 marks**, and **no appeals** will be entertained.
- You may use any programming language for assignments. Note that Excel is excluded and cannot be used for programming tasks.

Problems:

1.1 Object

The purpose of this assignment is to help you develop a clearer and more concrete understanding of what a market "bubble" looks like in terms of observable indicators.

1) **2015 episode**: June 19, 2014 to June 12, 2015
2) **2021 episode**: January 4, 2019 to February 18, 2021
3) **the current episode**: starting on September 18, 2024

1.2 Core Tasks and Suggested Output Style

Broadly, what you need to do is:

1) Compute **each indicator for the 2015 and 2021 episodes**.
2) Compute **the same indicator** for the **current** episode (from 2024-09-18 to 2026-03-09).
3) Compare the values across episodes and provide interpretive **discussions**.
4) Draw a **conclusion** about which **stage** do you think the current episode **arrives**.

You may write your results in a structure like the following (this is an example of the expected logic and formatting):

1) "In the 2015 bull market, Indicator A equals X."
2) "In the 2021 bull market, Indicator A equals Y."
3) "As of 2026-03-09, in the current bull market, Indicator A equals Z."
4) "So we can see in the episodes, Indicator A will increase / decrease / …"
5) "I think the current episode is in the early- / mid- / last-stage because …" (Combine all indicators together and draw a conclusion at the end of the report)

In addition, you need to plot time-series trend charts for the current episode: from the start of the current bull market (2024-09-18) to 2026-03-09, showing **the time trend of all indicators (calculate the indicators from start to each date before 2026-03-09).**

1.3 Indicators You Need to Compute (All Required)

1) **Trading Value Ratio**
   a) Compute the ratio: (A-share total trading value at bull market end) / (A-share total trading value at bull market start).
2) **Maximum Trading Value Ratio Within the Bull Market** (suppose the maximum trading value of the current episode has occurred)
   a) Compute the ratio: (Maximum A-share total trading value during the bull market interval) / (A-share total trading value at bull market start).
   b) Then, compare the SSE Composite Index (000001) performance before v.s. after the trading value peak: **compare the price change ($\frac{Price_{peak}}{Price_{start}} - 1$) before trading value reaches its peak versus the price change ($\frac{Price_{end}}{Price_{peak}} - 1$) after the peak** (within the bull market interval). Based on this comparison, what **conclusion** can you draw? You don't need to draw time trend picture for the Index perofomance.
3) **Margin Trading Balance Ratio**
   a) Compute the ratio: (Margin trading & securities lending balance at the bull market end) / (Margin trading & securities lending balance at the bull market start).
4) **New Issuance of Equity-Oriented Funds (Monthly)**
   a) Compute the ratio: (Newly established equity-oriented fund shares in the month of the bull market end) / (Newly established equity-oriented fund shares in the month

of the bull market start).

b) For the current episode (starting 2024-09-18), because the bull market end date has not occurred yet, **use February 2026 as the "end month"** for the purpose of this calculation.

c) If there is no value for the start month or the end month, just use the nearest value. For example, if you find that there is no newly established equity-oriented fund value in 2014m6, just use the value of 2014m7 (because this is the nearest month in the interval of 2015 episode).

5) **CSI 300 ERP (Equity Risk Premium)**: Level Change and Percentage Change During the Bull Market

a) Use CSI 300 (000300) to compute ERP during the bull market interval:

$$ERP = \frac{1}{CSI\ 300\ PE} - 10year\ government\ bond\ yield$$

b) For ERP, you must analyze both:

i. The change in ERP **values** during the bull market (start v.s. end).

ii. The change in ERP **percentiles**, constructed in the following way: because CSI 300 has been available since 2005, define a historical ERP distribution (**Empirical CDF**) using data that ends before the bubble period. For example, for the 2015 bull market, construct the ERP distribution using 2005 to 2013-12-31. Then evaluate where the 2015 bull market start ERP and end ERP fall in this distribution (their percentiles).

c) Remember to check the **unit** of 10-year government bond yield. Otherwise, all ERP values would be negative.

d) You don't need to draw the time-series figure of EPS percentile trend during the current episode.

6) **CSI 300 Dividend Yield Minus 10-Year Government Bond Yield**: Level and Percentile Changes

a) Compute (CSI 300 dividend yield – 10year government bond yield).

b) As with ERP, you need to analyze the **value changes** and the **percentile changes** during the bull market. Use the **same** method to define **distributions** as ERP.

c) You don't need to draw the time-series figure of the percentile trend during the current episode.

7) Price Change of Representative Stocks

a) Compute the price change over the bull market interval for **three** representative stocks.

i. A stock related to securities account opening: **East Money Information Co., Ltd. (300059.SZ)**

ii. A stock related to securities trading systems: **Hundsun Techologies Inc.**

(600570)

iii. A stock that benefits from higher trading value: brokerage firms – choose the **single brokerage stock with the largest price increase** in the interval, and report its increase

b) For the time-series trend figure of the current episode, you can just plot the price increase of East Money, Hundsun, and **the best-performance brokerage stock you find during the whole interval**. **You don't need to find each best-performance brokerage stock during the rolling windows.**

1.4 Data Sources

1) Data you can download from CSMAR

   You are required to download the following datasets from CSMAR:

   a) Daily total value of shares traded for the entire A-share market (covering Main board and GEM board) from *China Stock Market Series- Stock Trading* database.
   b) SSE Composite Index (000001) daily price from *China Stock Market Series- Market Index* database.
   c) Daily margin trading and securities lending balance from *China Stock Market Series – Margin Trading and Securities Lending* database.
   d) Daily 10-year government bond yield from *China Bond Market Series – Bond Market Trading – Interest Rate and Yield* database.

2) Data provided to you

   The following daily datasets have already been prepared for you:

   a) Daily newly established equity-oriented fund shares in "daily fund shares.dta".
   b) Daily CSI 300 PE and dividend yield in "CSI300 PE and dividend yield.dta".
   c) Daily prices for the representative stocks (East Money, Hundsun, and brokerage stocks) in "daily representative price.dta".

1.5 What Your Submission Must Include

You submission should include, at minimum:

1) The computed **indicator values for 2015, 2021, and the current episode**.
2) Clear cross-episode **comparisons and discussions**.
3) **Time-series trend plots** for all indicators from 2024-09-18 through 2026-03-09.
4) **Overall Assessment** of the Current Market Bubble.